SkeMex: Medical Reasoning with Self-Evolving Skill Memory

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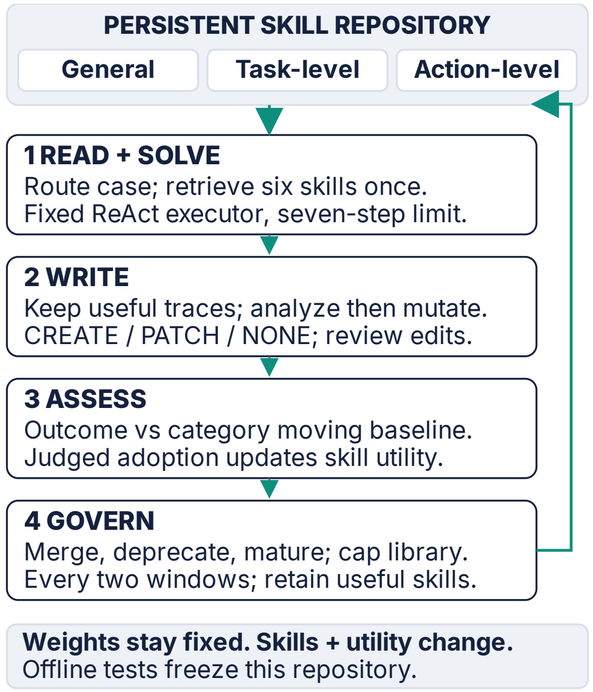
evidence: retained

Paper v4

Code

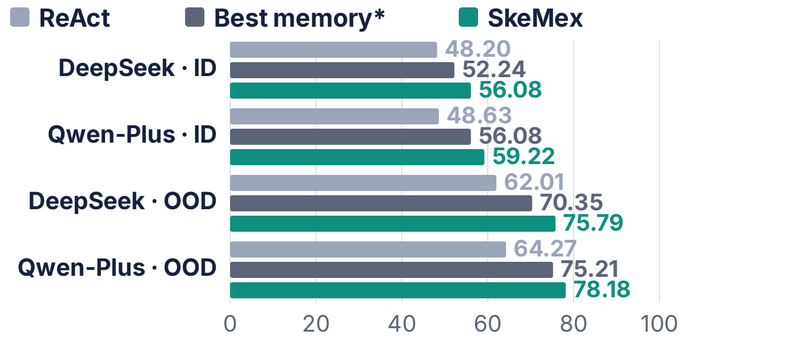
1 What changes and survives

Turn case experience into reusable procedures, not model-weight updates.



Content review + later utility governance, **not** a held-out regression gate per edit. The writer receives answer/rubric feedback on accumulation cases. §§3.1–3.6; App. F/L.

2 Frozen memory, held-out tests



Averages (%) · ID: 7 configurations, Table 1; OOD: 5, Table 14. *GSEM / CFM (ID), Evolver (OOD). No SD/CI or run count stated.

4 Uneven gains; real inference cost

- **Largest ReAct gains:** DeepSeek MMMU 46.04 → 66.91 (ID); AgentClinic-Text 34.11 → 68.22 (OOD). Tables 1/14.
- **Loss:** DeepSeek MedJourney 76.52 vs Evolver 78.79. LiveClin-MM ties Voyager/GSEM at 61.62 (ID).
- **Cost:** DeepSeek 4.77 steps / 116.06 s vs ReAct 3.17 / 54.48 s (Tables 16–17). Approximate inference time; memory-building cost excluded. No equal-total-compute test.

Table 14 used; Figure 3 differs (OOD baselines).

3 Controls and boundaries

Lifecycle ablation · online · Table 2

Five-config average	DeepSeek	Qwen-Plus
Full SkeMex	**53.22**	**56.19**
No skill writing	45.34	47.13
No utility valuation	47.16	49.32
No governance	50.02	52.50
Similarity-only retrieval	47.64	49.93

Same tools, prompts and inference limits per backbone; a different task mix/regime from the chart.

Frozen DeepSeek library → new consumers

Table 15	ReAct	SkeMex
Claude Sonnet-4.6	49.59	**60.27**
Qwen3.6-35B-A3B	47.71	**58.23**

Fixed skills shared; no target rewriting. Case history resets. Equal source construction budgets not established.

Offline ≠ online

Offline: train-only memory, no OOD accumulation; cross-split deduplication. Test traces excluded. No quantified contamination audit.

Online: scored traces can inform later memory. Epoch reuse/reset details unclear; not frozen held-out evidence (Table 3).

Baselines: same executor/split, own memory designs—not granted SkeMex governance.

Takeaway

Useful medical procedures can survive as a portable external memory. Frozen tests and lifecycle ablations support this package—not universal wins, clinical safety, or evolution of the improvement algorithm itself.

Hyperparameter-selection split / baseline tuning unclear. Independent validation unspecified; not evidence of test-set tuning. Original diagrams · arXiv 2606.09365v4 · details: blogs / SOURCE.md.

Our read · HarnessRSI

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Reports: frozen transfer, lifecycle controls, partial inference-budget matching.
Missing: run uncertainty, full construction cost, equal-total-compute test.